PROGRAM 1

Source code:

```
#include <iostream.h>
#include <conio.h>
Void main()
{
    int a,b;
    Cout << "Enter two numbers:";
    cin >> a>>b;
    Cout << "\n---Arithmetic operations---\n";
    Cout << "Addition:" << a+b << endl;
    Cout << "Subtraction:" << a-b << endl;
    Cout << "Multiplication:" << a*b << endl;
    if (b!=0)
    {
        Cout << "Division:" << a/b << endl;
        Cout << "Modulus:" << a%b << endl;
    }
    else
    {
        Cout << "Division and modulus by zero is not allowed";
    }
    return 0;
}
```

PROGRAM 2

Source code:

```
#include <iostream.h>
int main()
{
    double num1,num2;
    cout << "Enter first number:";
    cin >> num1;
    cout << "Enter Second number:";
    cin >> num2;
    if (num1>num2)
    {
        cout << "The biggest number is :" << num1 << endl;
    }
    else if (num2>num1)
    {
        cout << "The biggest number is :" << num2 << endl;
    }
    else
    {
        cout << "Both numbers are equal" << endl;
    }
    return 0;
}
```

PROGRAM 3

Source code:

```
#include <iostream.h>
#include <conio.h>
#include <string.h>
class login
{
    char Username[20];
    char Password[20];
    Public
    Void getData(char Username[], char Password[])
    {
        Cout << "Enter your Username:";
        cin >> Username;
        Cout << "Enter your Password:";
        cin >> Password;
    }
    int checklogin(char Username[], char Password[])
    {
        if (strcmp(Username,"admin")==0 && strcmp(Password,"admin@2005"))
            return 1;
        else
            return 2;
    }
};
void main()
{
    clrscr();
```

```
    login log;
    int attempts = 3;
    while (attempts > 0)
    {
        Cout << "Attempts left:" << attempts << endl;
        char Username[20], Password[20];
        log.getData(Username, Password);
        if (log.checklogin(Username, Password))
        {
            Cout << "login Successful";
            break;
        }
        else
        {
            Cout << "invalid login" << endl;
            attempts--;
        }
    }
    if (attempts == 0) {
        Cout << "login failed: No attempts left";
    }
    getch();
}
```

PROGRAM 4

Source code:

```
#include <iostream.h>
int main()
{
    int n;
    Cout << "Enter the length of array:";
    cin >> n;
    int arr[n];
    Cout << "Enter the elements of array:";
    for (int i=0; i<n; i++)
    {
        cin >> arr[i];
    }
    int maxval = arr[0];
    int minval = arr[0];
    for (int i=1; i<n; i++)
    {
        if (arr[i] > maxval)
        {
            maxval = arr[i];
        }
        else if (arr[i] < minval)
        {
            minval = arr[i];
        }
    }
    Cout << "Maximum Value:" << maxval << endl;
    Cout << "Minimum Value:" << minval << endl;
    return 0;
}
```

PROGRAM 5

Source code:

```
#include <iostream.h>
#include <conio.h>
class recursion()
{
public:
    int factorial(int n);
};

int recursion::factorial(int n)
{
    if (n==0)
        return 1;
    else
        return n * factorial(n-1);
}

int main()
{
    int number, result;
    Cout << "Enter Number:";
    cin >> number;
    recursion r;
    result = r.Factorial(number);
    Cout << result;
    return 0;
}
```

PROGRAM 6

Source code:

```
#include <iostream.h>
#include <conio.h>
class array_ex {
public:
    int add[10][10], A[10][10], B[10][10], i, j, row1, row2,
        col1, col2;
    void input();
};

void array_ex::input()
{
    clrscr();
    Cout << "Enter number of rows for matrix 1:";
    cin >> row1;
    Cout << "Enter number of Column for matrix 1:";
    cin >> col1;
    Cout << "Enter number of rows for matrix 2:";
    cin >> row2;
    Cout << "Enter number of column for matrix 2:";
    cin >> col2;
    if (row1==row2 && (col1==col2))
    {
        Cout << "Enter A matrix element:" << endl;
        for (i=0; i<row1; i++)
        {
            for (j=0; j<col1; j++)
            {
                cin >> A[i][j];
            }
        }
```

```
        Cout << endl;
    }
    Cout << "Enter B matrix element:" << endl;
    for (i=0; i<row2; i++)
    {
        for (j=0; j<col2; j++)
        {
            cin >> B[i][j];
        }
    }
    Cout << endl;
    Cout << "Addition of matrix:" << endl;
    for (i=0; i<row1; i++) {
        for (j=0; j<col1; j++) {
            add[i][j] = A[i][j] + B[i][j];
            Cout << add[i][j] << "\t";
        }
        Cout << endl;
    }
    }
    else
    {
        Cout << "Matrix does not Perform addition";
    }
}
int main()
{
```

```
    array_ex a;
    a.input();
    getch();
    return 0;
}
```

PROGRAM 7

Source code:

```
#include <iostream.h>
#include <string.h>
#include <algorithm.h>
class stringoperations
{
private:
    string str1, str2;
Public:
    void input();
    void length();
    void Concatenate();
    void Compare();
    void Copy();
    void reversestring();
};

void stringoperations::input()
{
    Cout << "Enter First String:";
    cin >> str1;
    Cout << "Enter second string:";
    cin >> str2;
}

void stringoperations::length()
{
    Cout << "\n length of first string:" << str1.length();
}

void stringoperations::Concatenate()
```

```
{
    string concat = str1 + " " + str2;
    Cout << "\n concatenated string:" << concat;
}

void stringoperations::compare()
{
    if (str1 == str2)
        Cout << "\n Both strings are equal.";
}

void stringoperations::Compare()
{
    if (str1 == str2)
        Cout << "\n Both strings are equal.";
    else
        Cout << "\n Strings are not equal.";
}

void stringoperations::copy()
{
    string copy = str1;
    Cout << "\n copied string:" << copy;
}

void stringoperations::reversestring()
{
    string rev = str1;
    reverse(rev.begin(), rev.end());
    Cout << "\n Reversed first string:" << rev;
}

int main()
```

```
{
    stringoperations s;
    s.input();
    s.length();
    s.Concatenate();
    s.Compare();
    s.Copy();
    s.reversestring();
    Cout << "\n\n Program executed Successfully.\n";
    return 0;
}
```